

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 12A

ENTROPY (SYSTEM, SURROUNDINGS & TOTAL ENTROPY)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 12: ENTROPY AND ENERGETICS
SUB-TOPIC:	12A: Entropy (System, Surroundings & Total Entropy)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	213 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Definition of System Entropy****[3 Marks]**
WCH14/01/Jan23/Q5

Entropy is a measure of the degree of disorder of a system.

(a) Define entropy, S , at a molecular level. [1]

(b) Predict, with a reason, whether entropy increases or decreases for: $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$. [2]

Q2. 2. Standard System Entropy Calculation: Haber Process**[4 Marks]**
WCH14/01/Oct22/Q6

Calculate ΔS_{system} for: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$

Standard entropies S° ($\text{J K}^{-1} \text{mol}^{-1}$): $\text{N}_2(\text{g})=191.6$, $\text{H}_2(\text{g})=130.6$, $\text{NH}_3(\text{g})=192.3$

(a) Calculate ΔS_{system} in $\text{J K}^{-1} \text{mol}^{-1}$. [3]

(b) Explain the sign of ΔS_{system} in terms of number of gas moles. [1]

Q3. 3. Surroundings Entropy Calculation**[3 Marks]**
WCH14/01/Jun22/Q7

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$, $\Delta H = -92.2 \text{ kJ mol}^{-1}$.

(a) Calculate $\Delta S_{\text{surroundings}}$ at 298 K using $\Delta S_{\text{surr}} = -\Delta H / T$. [2]

(b) State the units of $\Delta S_{\text{surroundings}}$. [1]

Q4. 4. Total Entropy and Feasibility Criterion**[3 Marks]**
WCH14/01/Jan22/Q8

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ at 298 K, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{mol}^{-1}$ and $\Delta S_{\text{surroundings}} = +309.4 \text{ J K}^{-1} \text{mol}^{-1}$.

(a) Calculate ΔS_{total} at 298 K. [1]

(b) State the criterion for a reaction to be spontaneous/feasible and deduce whether this reaction is spontaneous at 298 K. [2]

[4 Marks]

Q5. 5. Effect of Temperature on Feasibility

WCH14/01/Oct21/Q5

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$, $\Delta H = -92.2 \text{ kJ mol}^{-1}$ and $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate ΔS_{total} at 500 K. [2]

(b) Explain why increasing temperature makes this reaction less spontaneous. [2]

[4 Marks]

Q6. 6. Minimum Temperature of Feasibility Calculation

WCH14/01/Jun21/Q6

Thermal decomposition of $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$: $\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate the minimum temperature T_{min} at which this reaction becomes spontaneous ($\Delta S_{\text{total}} > 0$). [4]

[4 Marks]

Q7. 7. Entropy of Dissolution: Ammonium Nitrate

WCH14/01/Jan21/Q7

$\text{NH}_4\text{NO}_3(\text{s}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{NO}_3^-(\text{aq})$ is endothermic ($\Delta H = +25.7 \text{ kJ mol}^{-1}$) but dissolves spontaneously at 298 K.

(a) Explain in terms of ΔS_{system} why NH_4NO_3 dissolves spontaneously. [2]

(b) If $\Delta S_{\text{system}} = +108.7 \text{ J K}^{-1} \text{ mol}^{-1}$, calculate ΔS_{total} at 298 K. [2]

[3 Marks]

Q8. 8. Relating Total Entropy to Equilibrium Constant K

WCH14/01/Oct20/Q6

The total entropy change for a reaction is $\Delta S_{\text{total}} = R \ln K$.

(a) Calculate K when $\Delta S_{\text{total}} = +22.46 \text{ J K}^{-1} \text{ mol}^{-1}$ at 298 K. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) [2]

(b) State what $K > 1$ implies about the position of equilibrium. [1]

Q9. 9. Third Law of Thermodynamics & Absolute Zero Entropy

[2 Marks]

WCH14/01/Jun20/Q8

At 0 K, a perfect crystalline substance has $S = 0 \text{ J K}^{-1} \text{ mol}^{-1}$.

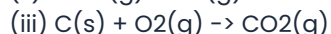
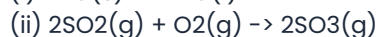
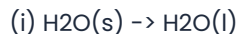
(a) Explain why entropy is zero at absolute zero (0 K). [2]

Q10. 10. Qualitative Entropy Changes Across States of Matter

[4 Marks]

WCH14/01/Jan20/Q9

Predict whether ΔS_{system} is positive or negative for:



(a) Give the sign of ΔS_{system} for each reaction with a brief reason. [4]

Q11. 11. Entropy of Combustion of Methane

[4 Marks]

WCH14/01/Oct19/Q10

$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O(l)}$: $\Delta H = -890 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -242 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate $\Delta S_{\text{surroundings}}$ at 298 K. [2]

(b) Calculate ΔS_{total} and explain why methane combustion is highly spontaneous. [2]

Q12. 12. Entropy vs Kinetic Inertness: Diamond to Graphite Conversion

[3 Marks]

WCH14/01/Jun19/Q11

$\text{C(diamond)} \rightarrow \text{C(graphite)}$: $\Delta H = -1.9 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +3.3 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate ΔS_{total} at 298 K and prove the conversion is thermodynamically spontaneous. [2]

(b) Explain why diamonds do not noticeably convert to graphite at room temperature. [1]

Q13. 13. Entropy of Evaporation of Water vs Ethanol

[4 Marks]
WCH14/01/Jan19/Q12

Trouton's rule states that $\Delta S_{\text{vapourisation}}$ is approximately +85 to +90 J K⁻¹ mol⁻¹ for many liquids.

(a) Explain why $\Delta S_{\text{vapourisation}}$ is similar for most simple liquids. [2]

(b) Explain why water has a higher $\Delta S_{\text{vapourisation}}$ (+109 J K⁻¹ mol⁻¹) than predicted by Trouton's rule. [2]

Q14. 14. Calculating Standard Entropies from Experimental Data Table

[4 Marks]
WCH14/01/sample/Q14

Reaction: $2\text{NaHCO}_3(\text{s}) \rightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$

S° values (J K⁻¹ mol⁻¹): $\text{NaHCO}_3(\text{s})=102$, $\text{Na}_2\text{CO}_3(\text{s})=136$, $\text{CO}_2(\text{g})=214$, $\text{H}_2\text{O}(\text{g})=189$

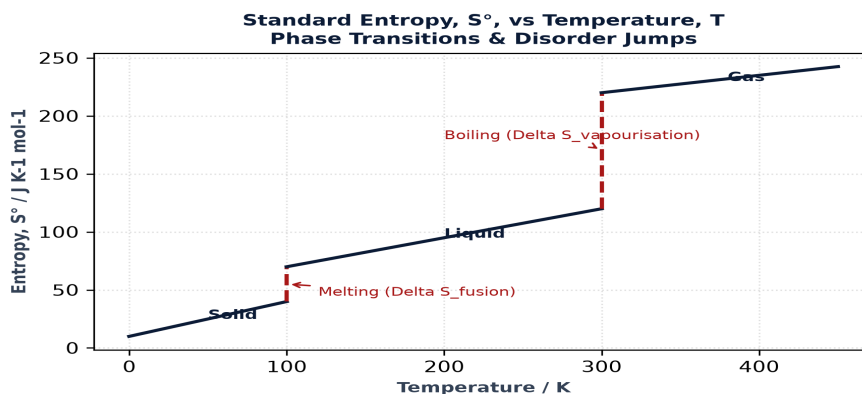
(a) Calculate ΔS_{system} . [3]

(b) Explain the large positive sign of ΔS_{system} . [1]

Q15. 15. Graphical Plot of Entropy S° vs Temperature T

[4 Marks]
WCH14/01/sample/Q15

A graph of standard entropy S° against temperature T from 0 K to 400 K shows two sharp vertical discontinuities.



(a) Identify what occurs at these two vertical jumps. [2]

(b) Explain why $\Delta S_{\text{vapourisation}}$ is larger than ΔS_{fusion} . [2]

Q16. 16. Endothermic Spontaneous Reactions

[3 Marks]

WCH14/01/Sample/Q16

$\text{Ba(OH)} \cdot 2.8\text{H}_2\text{O(s)} + 2\text{NH}_4\text{Cl(s)} \rightarrow \text{BaCl}_2\text{(s)} + 2\text{NH}_3\text{(g)} + 10\text{H}_2\text{O(l)}$ is strongly endothermic ($\Delta H = +164 \text{ kJ mol}^{-1}$).

(a) Explain how an endothermic reaction can occur spontaneously at room temperature. [3]

Q17. 17. Solubility Trends and Entropy of Hydration

[4 Marks]

WCH14/01/Sample/Q17

Dissolving MgSO_4 in water is exothermic ($\Delta H = -91 \text{ kJ mol}^{-1}$), whereas BaSO_4 is insoluble.

(a) Explain how cation charge density affects $\Delta S_{\text{hydration}}$ and ΔS_{system} . [2]

(b) Calculate $\Delta S_{\text{surroundings}}$ for MgSO_4 dissolution at 298 K. [2]

Q18. 18. Thermodynamic Equilibrium Constant Derivation from Entropy

[4 Marks]

WCH14/01/Sample/Q18

At equilibrium, $\Delta S_{\text{total}} = 0$ for the universe, leading to $\Delta S_{\text{total}} = R \ln K$.

(a) Deduce the expression for $\ln K$ in terms of ΔH and ΔS_{system} . [2]

(b) If $\Delta S_{\text{total}} = +50.0 \text{ J K}^{-1} \text{ mol}^{-1}$ at 298 K, calculate K . [2]

Q19. 19. Temperature Dependence of K via van 't Hoff and Entropy

[3 Marks]

WCH14/01/Sample/Q19

For an exothermic reaction ($\Delta H < 0$), explain how increasing temperature affects $\Delta S_{\text{surroundings}}$ and the equilibrium constant K .

(a) Relate $\Delta S_{\text{surroundings}} = -\Delta H / T$ to K using $\Delta S_{\text{total}} = R \ln K$. [3]

Q20. 20. Entropy Change in Gas Expansion

[3 Marks]

WCH14/01/Sample/Q20

1 mole of an ideal gas expands isothermally from 10 dm³ to 20 dm³ at 298 K.

(a) Calculate $\Delta S = R \ln(V_2/V_1)$. [2]

(b) Explain why expanding into a vacuum is spontaneous. [1]

Q21. 21. Entropy of Neutralisation Reaction

[3 Marks]

WCH14/01/Sample/Q21

$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$: $\Delta H = -57.1 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +80 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate ΔS_{total} at 298 K. [2]

(b) Explain why ΔS_{system} is positive when 2 ions form 1 liquid molecule. [1]

Q22. 22. Entropy Change of Rusting Iron

[4 Marks]

WCH14/01/Sample/Q22

$4\text{Fe}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{Fe}_2\text{O}_3(\text{s})$: $\Delta H = -1648 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -549 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]

(b) Explain why iron rusts spontaneously despite a large decrease in system entropy. [1]

Q23. 23. Entropy of Dehydration of Copper Sulfate Pentahydrate

[3 Marks]

WCH14/01/Sample/Q23

$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}) \rightarrow \text{CuSO}_4(\text{s}) + 5\text{H}_2\text{O}(\text{g})$: $\Delta H = +298 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +700 \text{ J K}^{-1} \text{ mol}^{-1}$.

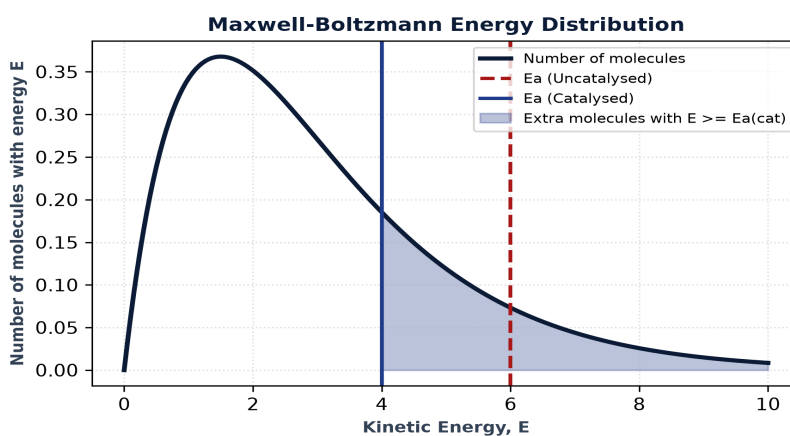
(a) Calculate the temperature at which $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ dehydrates spontaneously. [3]

Q24. 24. Microstates and Boltzmann Entropy Equation $S = k \ln W$

[3 Marks]

WCH14/01/Sample/Q24

Boltzmann's equation relates entropy to microstates: $S = k \ln W$.



(a) Explain what W represents. [1]

(b) Calculate S for a system with $W = 1.0 \times 10^{23}$ microstates. ($k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$) [2]

Q25. 25. Standard Entropy Comparison Across Halogen Elements

[3 Marks]

WCH14/01/Sample/Q25

S° values ($\text{J K}^{-1} \text{ mol}^{-1}$): $\text{F}_2(\text{g}) = 202.7$, $\text{Cl}_2(\text{g}) = 223.0$, $\text{Br}_2(\text{l}) = 152.2$, $\text{I}_2(\text{s}) = 116.1$.

(a) Explain why $\text{F}_2(\text{g})$ and $\text{Cl}_2(\text{g})$ have higher entropy than $\text{Br}_2(\text{l})$ and $\text{I}_2(\text{s})$. [2]

(b) Explain why $\text{Cl}_2(\text{g})$ has higher entropy than $\text{F}_2(\text{g})$. [1]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Complete Entropy Calculation & Equilibrium Constant K****[6 Marks]**
WCH14/01/Hard/Q26For the reaction: $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$ $\Delta H^\circ = -57.2 \text{ kJ mol}^{-1}$. $S^\circ (\text{J K}^{-1} \text{ mol}^{-1})$: $\text{NO}_2(\text{g}) = 240.0$, $\text{N}_2\text{O}_4(\text{g}) = 304.0$.(a) Calculate ΔS_{system} . [2]

(b) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [2]

(c) Calculate the equilibrium constant K at 298 K. [2]

Q27. 27. A* Challenge: Temperature at Which Equilibrium Constant K = 1**[5 Marks]**
WCH14/01/Hard/Q27For a reaction with $\Delta H = +125 \text{ kJ mol}^{-1}$ and $\Delta S_{\text{system}} = +160 \text{ J K}^{-1} \text{ mol}^{-1}$.(a) Show that when $K = 1$, $\Delta S_{\text{total}} = 0$. [1]

(b) Calculate the exact temperature in $^\circ\text{C}$ at which $K = 1$. [3]

(c) Calculate K at 1000 K. [1]

Q28. 28. A* Challenge: Thermodynamic Feasibility vs Rate Profile of Ammonia Synthesis**[5 Marks]**
WCH14/01/Hard/Q28Haber Process: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ $\Delta H = -92.2 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$.(a) Explain why industrial production uses 400–450 $^\circ\text{C}$ despite thermodynamics favouring low temperatures. [3]

(b) Calculate ΔS_{total} at 700 K and explain why high temperature prevents high equilibrium yield. [2]

Q29. 29. A* Challenge: Entropy of Solution and Lattice Energy Interplay

[5 Marks]
WCH14/01/Hard/Q29

Dissolving NaCl: $\text{NaCl(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$

$\Delta_{\text{LE}} H = -787 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}} H(\text{Na}^+) = -406 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}} H(\text{Cl}^-) = -364 \text{ kJ mol}^{-1}$.

Born-Haber Cycle for Sodium Chloride (NaCl)

_____	$\text{Na(g)} + \text{Cl(g)} + e^- (+725 \text{ kJ})$
_____	$\text{Na(g)} + e^- + \frac{1}{2} \text{Cl}_2(\text{g}) (+603 \text{ kJ})$
_____	$\text{Na(g)} + \text{Cl(g)} (+376 \text{ kJ})$
_____	$\text{Na(g)} + \frac{1}{2} \text{Cl}_2(\text{g}) (+107 \text{ kJ})$
_____	$\text{Na(s)} + \frac{1}{2} \text{Cl}_2(\text{g}) (+0 \text{ kJ})$
_____	$\text{NaCl(s)} (-411 \text{ kJ})$

(a) Calculate $\Delta_{\text{sol}} H$ for NaCl. [2]

(b) If $\Delta S_{\text{system}} = +43.0 \text{ J K}^{-1} \text{ mol}^{-1}$, calculate ΔS_{total} at 298 K and explain why NaCl dissolves. [3]

Q30. 30. A* Challenge: Temperature Feasibility Boundary Plot

[5 Marks]
WCH14/01/Hard/Q30

For $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO(s)} + \text{CO}_2(\text{g})$, $\Delta H = +178 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +161 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate ΔS_{total} at 800 K, 1105.6 K, and 1400 K. [3]

(b) Describe the graph of ΔS_{total} against temperature T. [2]

Q31. 31. A* Challenge: Industrial Extraction of Titanium via K and Entropy**[5 Marks]**
WCH14/01/Hard/Q31

Kroll Process: $\text{TiO}_2(\text{s}) + 2\text{C}(\text{s}) + 2\text{Cl}_2(\text{g}) \rightarrow \text{TiCl}_4(\text{l}) + 2\text{CO}(\text{g})$

$\Delta H = -230 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +295 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate ΔS_{total} at 1000 K. [2]

(b) Explain why both enthalpy and entropy drive this reaction at high temperature. [3]

Q32. 32. A* Challenge: Thermal Decomposition Trends of Group 2 Carbonates**[5 Marks]**
WCH14/01/Hard/Q32

Decomposition temperatures T_{decomp} : MgCO_3 (540 K), CaCO_3 (1110 K), SrCO_3 (1550 K), BaCO_3 (1630 K).

(a) Explain in terms of cation radius and polarizing power why T_{decomp} increases down Group 2. [3]

(b) Relate this trend to ΔH and $T_{\text{decomp}} = \Delta H / \Delta S_{\text{system}}$. [2]

Q33. 33. A* Challenge: Entropy of Polymerisation of Ethene**[5 Marks]**
WCH14/01/Hard/Q33

$n \text{ CH}_2=\text{CH}_2(\text{g}) \rightarrow -[\text{CH}_2-\text{CH}_2]_n(\text{s})$: $\Delta H = -105 \text{ kJ mol}^{-1}$ per monomer unit.

(a) Predict the sign of ΔS_{system} for addition polymerisation and explain why. [2]

(b) Calculate the maximum temperature T_{max} at which poly(ethene) is stable if $\Delta S_{\text{system}} = -70 \text{ J K}^{-1} \text{ mol}^{-1}$. [3]

Q34. 34. A* Challenge: Entropy of Protein Folding & Hydrophobic Effect**[5 Marks]**
WCH14/01/Hard/Q34

Unfolded polypeptide (random coil) → Folded native protein structure.

(a) Explain why polypeptide chain ordering gives negative ΔS_{chain} . [2]

(b) Explain how the hydrophobic effect (release of ordered water molecules) makes overall ΔS_{system} positive. [3]

Q35. 35. A* Challenge: Calculating ΔS_{total} for Cell Respiration**[5 Marks]**
WCH14/01/Hard/Q35

$\text{C}_6\text{H}_{12}\text{O}_6(\text{s}) + 6\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$: $\Delta H = -2803 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +259 \text{ J K}^{-1} \text{ mol}^{-1}$ at 310 K (37 °C).

(a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at body temperature (310 K). [3]

(b) Calculate the thermodynamic efficiency if 32 ATP molecules are formed ($\Delta G_{\text{ATP}} = +30.5 \text{ kJ mol}^{-1}$ each). [2]

Q36. 36. A* Challenge: Entropy Changes in Electrochemical Fuel Cells**[5 Marks]**
WCH14/01/Hard/Q36

In a hydrogen-oxygen fuel cell: $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$
 $\Delta H = -572 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -327 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate maximum electrical work $W_{\text{max}} = \Delta G = \Delta H - T \Delta S_{\text{system}}$ at 298 K. [3]

(b) Calculate theoretical cell emf E° using $\Delta G = -n F E^\circ$. ($n = 4$, $F = 96500 \text{ C mol}^{-1}$) [2]

Q37. 37. A* Challenge: Entropy of Vaporisation vs Hydrogen Bonding in Alcohols**[5 Marks]**
WCH14/01/Hard/Q37

S° data: Methanol CH₃OH(l)=127, CH₃OH(g)=240; Ethanol C₂H₅OH(l)=161, C₂H₅OH(g)=283.

(a) Calculate Delta S_{vap} for methanol and ethanol. [2]

(b) Compare with propane C₃H₈ (Delta S_{vap} = +87.5 J K⁻¹ mol⁻¹) and explain differences. [3]

Q38. 38. A* Challenge: Temperature Coefficient of Equilibrium Constant K**[5 Marks]**
WCH14/01/Hard/Q38

For N₂O₄(g) ⇌ 2NO₂(g), K = 0.144 at 298 K and K = 2.67 at 373 K.

(a) Calculate Delta S_{total} at 298 K and 373 K. [2]

(b) Deduce whether the forward reaction is endothermic or exothermic using Delta S_{surroundings} = -Delta H / T. [3]

Q39. 39. A* Challenge: Multi-Component Reaction Entropy Analysis**[5 Marks]**
WCH14/01/Hard/Q39

Solvay process step: Na₂CO₃(s) + CO₂(g) + H₂O(l) → 2NaHCO₃(s)
Delta H = -128 kJ mol⁻¹, Delta S_{system} = -335 J K⁻¹ mol⁻¹.

(a) Calculate Delta S_{total} at 298 K and state if reaction is spontaneous. [3]

(b) Calculate the temperature above which the reverse reaction (decomposition of NaHCO₃) becomes spontaneous. [2]

Q40. 40. A* Challenge: Gibbs-Helmholtz Relationship and Feasibility Plots**[5 Marks]**
WCH14/01/Hard/Q40

The Gibbs-Helmholtz equation is $\Delta G = \Delta H - T \Delta S_{\text{system}}$.

(a) Show that $\Delta S_{\text{total}} = -\Delta G / T$. [2]

(b) For a reaction with $\Delta H > 0$ and $\Delta S_{\text{system}} < 0$, explain why the reaction is unfeasible at ALL temperatures. [3]

Q41. 41. A* Challenge: Entropy of Sublimation of Iodine**[5 Marks]**
WCH14/01/Hard/Q41

$\text{I}_2(\text{s}) \rightarrow \text{I}_2(\text{g})$: $\Delta H_{\text{sub}} = +62.4 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +145 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate ΔS_{total} at 298 K. [2]

(b) Calculate the sublimation temperature of iodine at 1 atm pressure. [3]

Q42. 42. A* Challenge: Entropy of Mixing Ideal Gases**[5 Marks]**
WCH14/01/Hard/Q42

Mixing 1 mole of $\text{Ar}(\text{g})$ and 1 mole of $\text{Ne}(\text{g})$ at 298 K and 1 atm pressure.

(a) Calculate $\Delta S_{\text{mix}} = -R (n_{\text{A}} \ln x_{\text{A}} + n_{\text{B}} \ln x_{\text{B}})$ where $x_{\text{A}} = x_{\text{B}} = 0.5$. [3]

(b) Explain why mixing ideal gases involves $\Delta H_{\text{mix}} = 0$ but is strongly spontaneous. [2]

Q43. 43. A* Challenge: Entropy Analysis of Biological ATP Hydrolysis**[5 Marks]**
WCH14/01/Hard/Q43

$\text{ATP}^{4-}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{ADP}^{3-}(\text{aq}) + \text{HPO}_4^{2-}(\text{aq}) + \text{H}^+(\text{aq})$: $\Delta H = -20.5 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +34.0 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate ΔS_{total} at 310 K. [2]

(b) Calculate ΔG and equilibrium constant K at 310 K. [3]

Q44. 44. A* Challenge: High Temperature Thermal Cracking Entropy

[5 Marks]
WCH14/01/Hard/Q44

$\text{C}_{10}\text{H}_{22}(\text{g}) \rightarrow \text{C}_8\text{H}_{18}(\text{g}) + \text{C}_2\text{H}_4(\text{g})$: $\Delta H = +82 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +140 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate ΔS_{total} at 500 K and 900 K. [3]

(b) Explain why thermal cracking is carried out at high temperatures (700–900 K). [2]

Q45. 45. A* Challenge: Entropy of Hydration vs Ionic Radius Trend

[5 Marks]
WCH14/01/Hard/Q45

Hydration entropies ΔS_{hyd} ($\text{J K}^{-1} \text{ mol}^{-1}$): $\text{Li}^+ = -119$, $\text{Na}^+ = -89$, $\text{K}^+ = -67$, $\text{Rb}^+ = -52$.

(a) Explain the trend in ΔS_{hyd} down Group 1. [3]

(b) Relate this to ionic charge density and water orientation. [2]

Q46. 46. A* Challenge: Entropy of Combustion of Carbon Monoxide

[5 Marks]
WCH14/01/Hard/Q46

$2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g})$: $\Delta H = -566 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -173 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate ΔS_{total} at 298 K. [2]

(b) Calculate the upper temperature limit T_{max} above which CO combustion becomes unfeasible. [3]

Q47. 47. A* Challenge: Calculating System Entropy from Experimental S° Data

[5 Marks]
WCH14/01/Hard/Q47

For $\text{Fe}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{Fe}(\text{s}) + 3\text{H}_2\text{O}(\text{g})$

S° ($\text{J K}^{-1} \text{mol}^{-1}$): $\text{Fe}_2\text{O}_3(\text{s})=87.4$, $\text{H}_2(\text{g})=130.6$, $\text{Fe}(\text{s})=27.3$, $\text{H}_2\text{O}(\text{g})=188.7$.

(a) Calculate ΔS_{system} . [3]

(b) Explain why ΔS_{system} is positive despite 3 gas moles reacting to form 3 gas moles. [2]

Q48. 48. A* Challenge: Total Entropy vs Cell Potential E° Proof

[5 Marks]
WCH14/01/Hard/Q48

Prove that $\Delta S_{\text{total}} = n F E^\circ / T$.

(a) Combine $\Delta G = -n F E^\circ$ and $\Delta S_{\text{total}} = -\Delta G / T$ to derive the expression. [3]

(b) Calculate ΔS_{total} at 298 K for a cell with $E^\circ = +1.10 \text{ V}$ and $n = 2$. [2]

Q49. 49. A* Challenge: Entropy of Crystallisation of Hydrated Salts

[5 Marks]
WCH14/01/Hard/Q49

$\text{Cu}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 5\text{H}_2\text{O}(\text{l}) \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$: $\Delta H = -78 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = -210 \text{ J K}^{-1} \text{mol}^{-1}$.

(a) Calculate ΔS_{total} at 298 K. [2]

(b) Explain why crystallisation occurs spontaneously below 371 K. [3]

**Q50. 50. A* Challenge: Comprehensive Synthesis of
Thermodynamics & Kinetics**

[6 Marks]
WCH14/01/Hard/Q50

For $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g})$: $\Delta H = +180.6 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}} = +24.7 \text{ J K}^{-1} \text{ mol}^{-1}$, $E_a = +315 \text{ kJ mol}^{-1}$.

(a) Calculate ΔS_{total} at 298 K and 2000 K. [3]

(b) Explain why NO is not formed in air at 298 K due to BOTH thermodynamics and kinetics. [3]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 12A

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARK S
Q1	Q1: 1. Definition of System Entropy 1. (a) Measure of the number of ways of arranging energy/particles in a system (disorder) (1). 1. (b) Increases (1). Gas molecules have higher kinetic energy and far more random arrangements than liquid (1).	[3]
Q2	Q2: 2. Standard System Entropy Calculation: Haber Process 2. (a) $\sum S^\circ(\text{products}) = 2 \times 192.3 = 384.6$ (1). $\sum S^\circ(\text{reactants}) = 191.6 + 3(130.6) = 583.4$ (1). $\Delta S_{\text{sys}} = 384.6 - 583.4 = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 2. (b) Negative because 4 moles of gas react to form 2 moles of gas $\rightarrow$ decrease in disorder (1).	[4]
Q3	Q3: 3. Surroundings Entropy Calculation 3. (a) $\Delta S_{\text{surr}} = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 3. (b) $\text{J K}^{-1} \text{ mol}^{-1}$ (1).	[3]
Q4	Q4: 4. Total Entropy and Feasibility Criterion 4. (a) $\Delta S_{\text{total}} = -198.8 + 309.4 = +110.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 4. (b) Spontaneous when $\Delta S_{\text{total}} > 0$ (1). Yes, feasible because $+110.6 > 0$ (1).	[3]
Q5	Q5: 5. Effect of Temperature on Feasibility 5. (a) At 500 K: $\Delta S_{\text{surr}} = -(-92200) / 500 = +184.4 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = -198.8 + 184.4 = -14.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 5. (b) As T increases, $\Delta S_{\text{surroundings}} (-\Delta H / T)$ becomes less positive, making ΔS_{total} negative (2).	[4]
Q6	Q6: 6. Minimum Temperature of Feasibility Calculation 6. (a) At feasibility boundary: $\Delta S_{\text{total}} = 0 \Rightarrow \Delta S_{\text{surroundings}} = -\Delta S_{\text{system}} = -161 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). $-\Delta H / T = -161 \Rightarrow -178000 / T = -161$ (1). $T = 178000 / 161 = 1105.6 \text{ K}$ (2).	[4]
Q7	Q7: 7. Entropy of Dissolution: Ammonium Nitrate 7. (a) Solid ionic lattice breaks down into free mobile hydrated ions, creating a large increase in disorder ($\Delta S_{\text{system}} > 0$) (2). 7. (b) $\Delta S_{\text{surr}} = -25700 / 298 = -86.24 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = +108.7 - 86.24 = +22.46 \text{ J K}^{-1} \text{ mol}^{-1}$ (2).	[4]
Q8	Q8: 8. Relating Total Entropy to Equilibrium Constant K 8. (a) $\ln K = \Delta S_{\text{total}} / R = 22.46 / 8.31 = 2.7028$ (1). $K = e^{2.7028} = 14.9$ (1). 8. (b) Equilibrium lies predominantly to the right (products favoured) (1).	[3]
Q9	Q9: 9. Third Law of Thermodynamics & Absolute Zero Entropy 9. (a) At 0 K, all molecular translational, rotational, and vibrational motion ceases (1). There is only 1 microstate / zero disorder (1).	[2]
Q10	Q10: 10. Qualitative Entropy Changes Across States of Matter 10. (a) (i) Positive (solid to liquid) (1). (ii) Negative (3 mol gas to 2 mol gas) (1). (iii) Approximately zero / slightly positive (1 mol gas to 1 mol gas) (1). (iv) Positive (lattice breaks into mobile hydrated ions) (1).	[4]
Q11	Q11: 11. Entropy of Combustion of Methane 11. (a) $\Delta S_{\text{surr}} = -(-890000) / 298 = +2986.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 11. (b) $\Delta S_{\text{total}} = -242 + 2986.6 = +2744.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Extremely positive ΔS_{total} makes combustion overwhelming spontaneous (1).	[4]
Q12	Q12: 12. Entropy vs Kinetic Inertness: Diamond to Graphite Conversion 12. (a) $\Delta S_{\text{surr}} = -(-1900) / 298 = +6.38 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = +3.3 + 6.38 = +9.68 \text{ J K}^{-1} \text{ mol}^{-1} > 0$ (spontaneous) (2). 12. (b) High activation energy due to breaking strong covalent C-C bonds makes the reaction kinetically inert (1).	[3]

Q13	Q13: 13. Entropy of Evaporation of Water vs Ethanol 13. (a) Boiling represents transition from a disordered liquid to an ideal gas with similar disorder increase for 1 mole (2). 13. (b) Liquid water has an unusually ordered hydrogen-bonded network. Breaking this ordered structure requires extra entropy gain (2).	[4]
Q14	Q14: 14. Calculating Standard Entropies from Experimental Data Table 14. (a) $\sum S^\circ(\text{products}) = 136 + 214 + 189 = 539 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). $\sum S^\circ(\text{reactants}) = 2 \times 102 = 204 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). $\Delta S_{\text{sys}} = 539 - 204 = +335 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 14. (b) 2 moles of solid produce 1 mole of solid and 2 moles of gas (2 gas moles formed) (1).	[4]
Q15	Q15: 15. Graphical Plot of Entropy S° vs Temperature T 15. (a) Melting point (solid to liquid) and boiling point (liquid to gas) phase changes (2). 15. (b) Gas phase has vastly greater disorder and volume expansion than liquid phase compared to minor disorder change from solid to liquid (2).	[4]
Q16	Q16: 16. Endothermic Spontaneous Reactions 16. (a) 3 moles of solid produce 1 mole of solid, 2 moles of gas, and 10 moles of liquid (huge increase in particles/disorder) (1). ΔS_{system} is extremely positive (1). $\Delta S_{\text{system}} > \Delta S_{\text{surroundings}} $, so $\Delta S_{\text{total}} > 0$ (1).	[3]
Q17	Q17: 17. Solubility Trends and Entropy of Hydration 17. (a) Mg^{2+} has higher charge density than Ba^{2+} , causing stronger ordering of surrounding water molecules ($-\Delta S_{\text{hyd}}$) (2). 17. (b) $\Delta S_{\text{surr}} = -(-91000) / 298 = +305.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2).	[4]
Q18	Q18: 18. Thermodynamic Equilibrium Constant Derivation from Entropy 18. (a) $\Delta S_{\text{total}} = \Delta S_{\text{system}} - \Delta H / T = R \ln K \Rightarrow \ln K = \Delta S_{\text{system}} / R - \Delta H / (RT)$ (2). 18. (b) $\ln K = 50.0 / 8.31 = 6.0169 \Rightarrow K = e^{6.0169} = 410.3$ (2).	[4]
Q19	Q19: 19. Temperature Dependence of K via van 't Hoff and Entropy 19. (a) As T increases, $\Delta S_{\text{surroundings}} (-\Delta H / T)$ becomes less positive (1). ΔS_{total} decreases (1). Since $\ln K = \Delta S_{\text{total}} / R$, K decreases (equilibrium shifts left) (1).	[3]
Q20	Q20: 20. Entropy Change in Gas Expansion 20. (a) $\Delta S = 8.31 \times \ln(20 / 10) = 8.31 \times 0.69315 = +5.76 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 20. (b) Gas molecules have double the volume/microstates available, increasing disorder (1).	[3]
Q21	Q21: 21. Entropy of Neutralisation Reaction 21. (a) $\Delta S_{\text{surr}} = -(-57100) / 298 = +191.6 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = 80 + 191.6 = +271.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 21. (b) H^+ and OH^- ions strongly order surrounding water molecules; forming H_2O releases ordered hydration shells (1).	[3]
Q22	Q22: 22. Entropy Change of Rusting Iron 22. (a) $\Delta S_{\text{surr}} = -(-1648000) / 298 = +5530.2 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{total}} = -549 + 5530.2 = +4981.2 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 22. (b) The huge exothermic heat release creates an overwhelming positive surroundings entropy change (1).	[4]
Q23	Q23: 23. Entropy of Dehydration of Copper Sulfate Pentahydrate 23. (a) $T_{\text{min}} = \Delta H / \Delta S_{\text{system}} = 298000 \text{ J mol}^{-1} / 700 \text{ J K}^{-1} \text{ mol}^{-1} = 425.7 \text{ K}$ (152.6 °C) (3).	[3]
Q24	Q24: 24. Microstates and Boltzmann Entropy Equation $S = k \ln W$ 24. (a) W = number of energetically equivalent microstates / arrangements (1). 24. (b) $S = (1.38 \times 10^{-23}) \times \ln(1.0 \times 10^{23}) = (1.38 \times 10^{-23}) \times 52.956 = 7.31 \times 10^{-22} \text{ J K}^{-1}$ (2).	[3]
Q25	Q25: 25. Standard Entropy Comparison Across Halogen Elements 25. (a) Gases have far higher molecular kinetic motion and freedom of movement than liquids or solids (2). 25. (b) Cl_2 has more electrons / higher molar mass, giving more vibrational and rotational energy levels (1).	[3]

Q26	Q26: 26. A* Challenge: Complete Entropy Calculation & Equilibrium Constant K 26. (a) $\Delta S_{\text{sys}} = 304.0 - 2(240.0) = -176.0 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 26. (b) $\Delta S_{\text{surr}} = -(-57200)/298 = +191.95 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = -176.0 + 191.95 = +15.95 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 26. (c) $\ln K = 15.95 / 8.31 = 1.9194 \Rightarrow K = e^{1.9194} = 6.82$ (2).	[6]
Q27	Q27: 27. A* Challenge: Temperature at Which Equilibrium Constant K = 1 27. (a) $\Delta S_{\text{total}} = R \ln K = R \ln(1) = 0 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 27. (b) $T = \Delta H / \Delta S_{\text{sys}} = 125000 / 160 = 781.25 \text{ K} = 508.1^\circ\text{C}$ (3). 27. (c) At 1000 K, $\Delta S_{\text{surr}} = -125000/1000 = -125$. $\Delta S_{\text{tot}} = 160 - 125 = +35$. $\ln K = 35/8.31 = 4.211 \Rightarrow K = 67.4$ (1).	[5]
Q28	Q28: 28. A* Challenge: Thermodynamic Feasibility vs Rate Profile of Ammonia Synthesis 28. (a) Low temperature gives high equilibrium yield ($\Delta S_{\text{total}} > 0$) but reaction rate is too slow (high activation energy) (2). 400–450 °C is a compromise temperature giving acceptable rate with catalyst (1). 28. (b) At 700 K: $\Delta S_{\text{surr}} = -(-92200)/700 = +131.7 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = -198.8 + 131.7 = -67.1 \text{ J K}^{-1} \text{ mol}^{-1} < 0$ (unfeasible/low yield) (2).	[5]
Q29	Q29: 29. A* Challenge: Entropy of Solution and Lattice Energy Interplay 29. (a) $\Delta_{\text{sol}} H = \Delta_{\text{hyd}} H(\text{Na}^+) + \Delta_{\text{hyd}} H(\text{Cl}^-) - \Delta_{\text{LE}} H = (-406 - 364) - (-787) = +17.0 \text{ kJ mol}^{-1}$ (2). 29. (b) $\Delta S_{\text{surr}} = -17000 / 298 = -57.05 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = +43.0 - 57.05 = -14.05 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). Dissolves due to local concentration/entropy effects (1).	[5]
Q30	Q30: 30. A* Challenge: Temperature Feasibility Boundary Plot 30. (a) At 800 K: $\Delta S_{\text{surr}} = -178000/800 = -222.5 \Rightarrow \Delta S_{\text{tot}} = -61.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). At 1105.6 K: $\Delta S_{\text{tot}} = 0$ (1). At 1400 K: $\Delta S_{\text{surr}} = -127.1 \Rightarrow \Delta S_{\text{tot}} = +33.9 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 30. (b) Curve increases monotonically with increasing T, crossing from negative to positive at $T = 1105.6 \text{ K}$ (2).	[5]
Q31	Q31: 31. A* Challenge: Industrial Extraction of Titanium via K and Entropy 31. (a) $\Delta S_{\text{surr}} = -(-230000)/1000 = +230 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = +295 + 230 = +525 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 31. (b) Exothermic ($\Delta H < 0 \Rightarrow \Delta S_{\text{surr}} > 0$) AND gas moles increase from 2 mol Cl_2 to 2 mol CO gas plus liquid TiCl_4 ($\Delta S_{\text{sys}} > 0$). Both terms are positive at all T (3).	[5]
Q32	Q32: 32. A* Challenge: Thermal Decomposition Trends of Group 2 Carbonates 32. (a) Cation radius increases down Group 2 ($\text{Mg}^{2+} < \text{Ca}^{2+} < \text{Sr}^{2+} < \text{Ba}^{2+}$) (1); charge density and polarizing power decrease (1); less distortion of carbonate C–O bond $\Rightarrow$ lattice more thermally stable (1). 32. (b) ΔH becomes more endothermic (more positive) down Group 2, increasing $T_{\text{decomp}} = \Delta H / \Delta S_{\text{system}}$ (2).	[5]
Q33	Q33: 33. A* Challenge: Entropy of Polymerisation of Ethene 33. (a) ΔS_{system} is negative because n moles of free gas monomers form 1 mole of rigid solid polymer chain (2). 33. (b) $T_{\text{max}} = \Delta H / \Delta S_{\text{sys}} = 105000 \text{ J mol}^{-1} / 70 \text{ J K}^{-1} \text{ mol}^{-1} = 1500 \text{ K}$ (1227 °C) (3). Above this temp, polymer unzips into monomer.	[5]
Q34	Q34: 34. A* Challenge: Entropy of Protein Folding & Hydrophobic Effect 34. (a) Folding restricts conformational freedom of the backbone and side chains, reducing microstates ($\Delta S_{\text{chain}} < 0$) (2). 34. (b) Non-polar side chains force surrounding water into ordered clathrate cages. When hydrophobic core buries side chains, ordered water is released into bulk water, giving a large positive $\Delta S_{\text{water}} > \Delta S_{\text{chain}} $ (3).	[5]
Q35	Q35: 35. A* Challenge: Calculating ΔS_{total} for Cell Respiration 35. (a) $\Delta S_{\text{surr}} = -(-2803000)/310 = +9041.9 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{total}} = 259 + 9041.9 = +9300.9 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 35. (b) Energy captured = $32 \times 30.5 = 976 \text{ kJ mol}^{-1}$. Efficiency = $(976 / 2803) \times 100\% = 34.8\%$ (2).	[5]

Q36	Q36: 36. A* Challenge: Entropy Changes in Electrochemical Fuel Cells 36. (a) $\Delta G = -572000 - 298(-327) = -572000 + 97446 = -474554 \text{ J mol}^{-1} = -474.6 \text{ kJ mol}^{-1}$ (3). 36. (b) $E^\circ = -(-474554) / (4 \times 96500) = 474554 / 386000 = +1.23 \text{ V}$ (2).	[5]
Q37	Q37: 37. A* Challenge: Entropy of Vaporisation vs Hydrogen Bonding in Alcohols 37. (a) Methanol: $240 - 127 = +113 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). Ethanol: $283 - 161 = +122 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 37. (b) Alcohols have hydrogen bonding in liquid state, increasing liquid order compared to non-polar propane. Breaking H-bonds requires additional entropy increase upon vaporisation (3).	[5]
Q38	Q38: 38. A* Challenge: Temperature Coefficient of Equilibrium Constant K 38. (a) At 298 K: $\Delta S_{\text{tot}} = 8.31 \ln(0.144) = -16.1 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). At 373 K: $\Delta S_{\text{tot}} = 8.31 \ln(2.67) = +8.16 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 38. (b) As T increases, K increases $\Rightarrow \Delta S_{\text{total}}$ becomes more positive $\Rightarrow \Delta S_{\text{surroundings}} (-\Delta H / T)$ becomes less negative $\Rightarrow \Delta H > 0$ (Endothermic) (3).	[5]
Q39	Q39: 39. A* Challenge: Multi-Component Reaction Entropy Analysis 39. (a) $\Delta S_{\text{surr}} = -(-128000)/298 = +429.5 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = -335 + 429.5 = +94.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (spontaneous) (3). 39. (b) For reverse reaction: $\Delta H_{\text{rev}} = +128 \text{ kJ mol}^{-1}$, $\Delta S_{\text{sys_rev}} = +335 \text{ J K}^{-1} \text{ mol}^{-1}$. $T_{\text{decomp}} = 128000 / 335 = 382.1 \text{ K}$ (109 °C) (2).	[5]
Q40	Q40: 40. A* Challenge: Gibbs-Helmholtz Relationship and Feasibility Plots 40. (a) $\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = \Delta S_{\text{sys}} - \Delta H / T = -(\Delta H - T \Delta S_{\text{sys}}) / T = -\Delta G / T$ (2). 40. (b) If $\Delta H > 0$, $\Delta S_{\text{surr}} (-\Delta H / T)$ is negative at all T. If $\Delta S_{\text{sys}} < 0$, both terms are negative, so ΔS_{total} is negative (ΔG positive) at all temperatures (3).	[5]
Q41	Q41: 41. A* Challenge: Entropy of Sublimation of Iodine 41. (a) $\Delta S_{\text{surr}} = -62400 / 298 = -209.4 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = +145 - 209.4 = -64.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 41. (b) $T_{\text{sub}} = \Delta H_{\text{sub}} / \Delta S_{\text{sys}} = 62400 / 145 = 430.3 \text{ K}$ (157.2 °C) (3).	[5]
Q42	Q42: 42. A* Challenge: Entropy of Mixing Ideal Gases 42. (a) $x_{\text{Ar}} = 0.5$, $x_{\text{Ne}} = 0.5$. $\Delta S_{\text{mix}} = -8.31 \times (1 \times \ln 0.5 + 1 \times \ln 0.5) = -8.31 \times 2 \times (-0.69315) = +11.52 \text{ J K}^{-1}$ (3). 42. (b) Ideal gas molecules have zero intermolecular forces so $\Delta H_{\text{mix}} = 0$ ($\Delta S_{\text{surr}} = 0$). Spontaneity is driven entirely by system entropy gain $\Delta S_{\text{mix}} > 0$ (2).	[5]
Q43	Q43: 43. A* Challenge: Entropy Analysis of Biological ATP Hydrolysis 43. (a) $\Delta S_{\text{surr}} = -(-20500)/310 = +66.13 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = 34.0 + 66.13 = +100.13 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 43. (b) $\Delta G = -T \Delta S_{\text{tot}} = -310 \times 100.13 = -31040 \text{ J mol}^{-1} = -31.04 \text{ kJ mol}^{-1}$ (1). $\ln K = 100.13 / 8.31 = 12.049 \Rightarrow K = e^{12.049} = 1.71 \times 10^5$ (2).	[5]
Q44	Q44: 44. A* Challenge: High Temperature Thermal Cracking Entropy 44. (a) At 500 K: $\Delta S_{\text{surr}} = -82000/500 = -164 \Rightarrow \Delta S_{\text{tot}} = 140 - 164 = -24 \text{ J K}^{-1} \text{ mol}^{-1}$ (unfeasible) (1.5). At 900 K: $\Delta S_{\text{surr}} = -82000/900 = -91.1 \Rightarrow \Delta S_{\text{tot}} = 140 - 91.1 = +48.9 \text{ J K}^{-1} \text{ mol}^{-1}$ (feasible) (1.5). 44. (b) High temp is required both to make $\Delta S_{\text{total}} > 0$ (thermodynamics) and overcome high C-C bond activation energy (kinetics) (2).	[5]
Q45	Q45: 45. A* Challenge: Entropy of Hydration vs Ionic Radius Trend 45. (a) ΔS_{hyd} becomes less negative down Group 1 as ionic radius increases (3). 45. (b) Li^+ has highest charge density, forming the most tightly ordered hydration shell (-s). Larger ions (K^+ , Rb^+) have lower charge density and organize surrounding water less strongly (2).	[5]
Q46	Q46: 46. A* Challenge: Entropy of Combustion of Carbon Monoxide 46. (a) $\Delta S_{\text{surr}} = -(-566000)/298 = +1899.3 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = -173 + 1899.3 = +1726.3 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 46. (b) $T_{\text{max}} = \Delta H / \Delta S_{\text{sys}} = 566000 \text{ J mol}^{-1} / 173 \text{ J K}^{-1} \text{ mol}^{-1} = 3271.7 \text{ K}$ (2998.5 °C) (3).	[5]

Q47	Q47: 47. A* Challenge: Calculating System Entropy from Experimental S° Data 47. (a) $\sum S^\circ(\text{products}) = 2(27.3) + 3(188.7) = 54.6 + 566.1 = 620.7$ (1). $\sum S^\circ(\text{reactants}) = 87.4 + 3(130.6) = 479.2$ (1). $\Delta S_{\text{sys}} = 620.7 - 479.2 = +141.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 47. (b) H ₂ O(g) has higher molar entropy (188.7) than H ₂ (g) (130.6) due to more rotational/vibrational modes, and Fe(s) is formed (2).	[5]
Q48	Q48: 48. A* Challenge: Total Entropy vs Cell Potential E° Proof 48. (a) $\Delta S_{\text{total}} = -\Delta G / T$ (1). Substitute $\Delta G = -n F E^\circ \Rightarrow \Delta S_{\text{total}} = -(-n F E^\circ) / T = n F E^\circ / T$ (2). 48. (b) $\Delta S_{\text{total}} = (2 \times 96500 \times 1.10) / 298 = 212300 / 298 = +712.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2).	[5]
Q49	Q49: 49. A* Challenge: Entropy of Crystallisation of Hydrated Salts 49. (a) $\Delta S_{\text{surr}} = -(-78000)/298 = +261.7 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = -210 + 261.7 = +51.7 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 49. (b) Exothermic heat release (+S _{surr}) outweighs system entropy decrease (-S _{sys}) at temperatures below $T = 78000 / 210 = 371.4 \text{ K}$ (3).	[5]
Q50	Q50: 50. A* Challenge: Comprehensive Synthesis of Thermodynamics & Kinetics 50. (a) At 298 K: $\Delta S_{\text{surr}} = -180600/298 = -606.0 \Rightarrow \Delta S_{\text{tot}} = +24.7 - 606.0 = -581.3 \text{ J K}^{-1} \text{ mol}^{-1}$ (unfeasible) (1.5). At 2000 K: $\Delta S_{\text{surr}} = -180600/2000 = -90.3 \Rightarrow \Delta S_{\text{tot}} = +24.7 - 90.3 = -65.6 \text{ J K}^{-1} \text{ mol}^{-1}$ (1.5). 50. (b) Thermodynamically unfeasible at 298 K ($\Delta S_{\text{total}} < 0$) AND kinetically inert due to high activation energy (+315 kJ mol ⁻¹ for N=N triple bond cleavage) (3).	[6]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 12A

FAQ #1: Units of System Entropy vs Enthalpy

Category: Units Conversion

COMMON EXAMINER TRAP / MISTAKE:

Adding ΔS_{system} ($\text{J K}^{-1} \text{mol}^{-1}$) directly to ΔH (kJ mol^{-1}).

MODEL ANSWER / EXAMINER EXPECTATION:

Always convert ΔH to Joules ($\times 1000$) before calculating $\Delta S_{\text{surroundings}} = -\Delta H / T$.
System and surroundings entropy MUST both be in $\text{J K}^{-1} \text{mol}^{-1}$.

FAQ #2: Minus Sign in Surroundings Entropy Formula

Category: Formula Trap

COMMON EXAMINER TRAP / MISTAKE:

Forgetting the minus sign in $\Delta S_{\text{surroundings}} = -\Delta H / T$.

MODEL ANSWER / EXAMINER EXPECTATION:

Exothermic reactions (ΔH negative) lose heat to surroundings $\rightarrow$ surroundings entropy INCREASES (ΔS_{surr} positive). The minus sign ensures this.

FAQ #3: Spontaneity Criterion

Category: Feasibility

COMMON EXAMINER TRAP / MISTAKE:

Stating $\Delta S_{\text{system}} > 0$ is required for spontaneity.

MODEL ANSWER / EXAMINER EXPECTATION:

Spontaneity depends ONLY on TOTAL entropy: $\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$. An endothermic reaction with negative ΔS_{sys} can still be spontaneous if surroundings entropy is highly positive.

FAQ #4: Temperature of Feasibility Calculation

Category: Threshold Calculation

COMMON EXAMINER TRAP / MISTAKE:

Using $\Delta S_{\text{surroundings}} = \Delta S_{\text{system}}$ at boundary.

MODEL ANSWER / EXAMINER EXPECTATION:

At the threshold of feasibility, $\Delta S_{\text{total}} = 0$, so $\Delta S_{\text{surroundings}} = -\Delta S_{\text{system}}$.
Therefore $-\Delta H / T_{\text{min}} = -\Delta S_{\text{sys}} \Rightarrow T_{\text{min}} = \Delta H / \Delta S_{\text{sys}}$.

FAQ #5: Entropy at Absolute Zero

Category: Third Law

COMMON EXAMINER TRAP / MISTAKE:

Assuming non-zero entropy at 0 K for perfect crystals.

MODEL ANSWER / EXAMINER EXPECTATION:

Third Law of Thermodynamics: The entropy of a pure, perfectly crystalline substance is EXACTLY ZERO at absolute zero (0 K).

FAQ #6: State Changes and Entropy

Category: Qualitative Entropy

COMMON EXAMINER TRAP / MISTAKE:

Predicting entropy decrease during melting or boiling.

MODEL ANSWER / EXAMINER EXPECTATION:

Melting (solid $\rightarrow$ liquid) and boiling (liquid $\rightarrow$ gas) ALWAYS increase system entropy due to increased molecular motion and disorder.

FAQ #7: Gas Moles and System Entropy Sign**Category: Qualitative Rules****COMMON EXAMINER TRAP / MISTAKE:**

Ignoring change in number of gas moles when predicting sign of ΔS_{system} .

MODEL ANSWER / EXAMINER EXPECTATION:

If gas moles INCREASE (reactants $\rightarrow$ products), ΔS_{system} is POSITIVE. If gas moles DECREASE, ΔS_{system} is NEGATIVE.

FAQ #8: Dissolution of Salts and Entropy**Category: Solution Entropy****COMMON EXAMINER TRAP / MISTAKE:**

Assuming dissolving an ionic lattice always increases entropy.

MODEL ANSWER / EXAMINER EXPECTATION:

Dissolving involves lattice breakdown (+S) AND ion hydration (-S due to water ordering). If hydration ordering dominates, ΔS_{system} can be negative.

FAQ #9: Kinetic Inertness vs Thermodynamic Feasibility**Category: Kinetics vs Thermodynamics****COMMON EXAMINER TRAP / MISTAKE:**

Assuming $\Delta S_{\text{total}} > 0$ guarantees a rapid reaction.

MODEL ANSWER / EXAMINER EXPECTATION:

$\Delta S_{\text{total}} > 0$ proves a reaction is THERMODYNAMICALLY FEASIBLE, but high activation energy (E_a) can make it kinetically inert (infinitely slow at room temp).

FAQ #10: Relating ΔS_{total} to Equilibrium Constant K**Category: Thermodynamic Equilibrium****COMMON EXAMINER TRAP / MISTAKE:**

Confusing R value or units in $\Delta S_{\text{total}} = R \ln K$.

MODEL ANSWER / EXAMINER EXPECTATION:

$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$. ΔS_{total} MUST be in $\text{J K}^{-1} \text{ mol}^{-1}$. $\ln K = \Delta S_{\text{total}} / R$. Exponentiate: $K = e^{(\Delta S_{\text{total}} / R)}$.